KP04 Three-Layer Perceptron: Detect Each Code, Then Merge

> Level: undergraduate, year 3
>
> Pattern Recognition | Chapter 4, §4.3.2

Slides: read and download

Learning objectives

- Write a detector for any three-bit target code and explain why its score is $0.5-$ (number of wrong bits).
- Join several detectors with an OR merge in the output layer and check the prediction at every code.
- Explain that the three-layer construction only proves that this is enough, not that the weights are learned or the network is minimal.

KP04 Three-Layer Perceptron: Detect Each Code, Then Merge

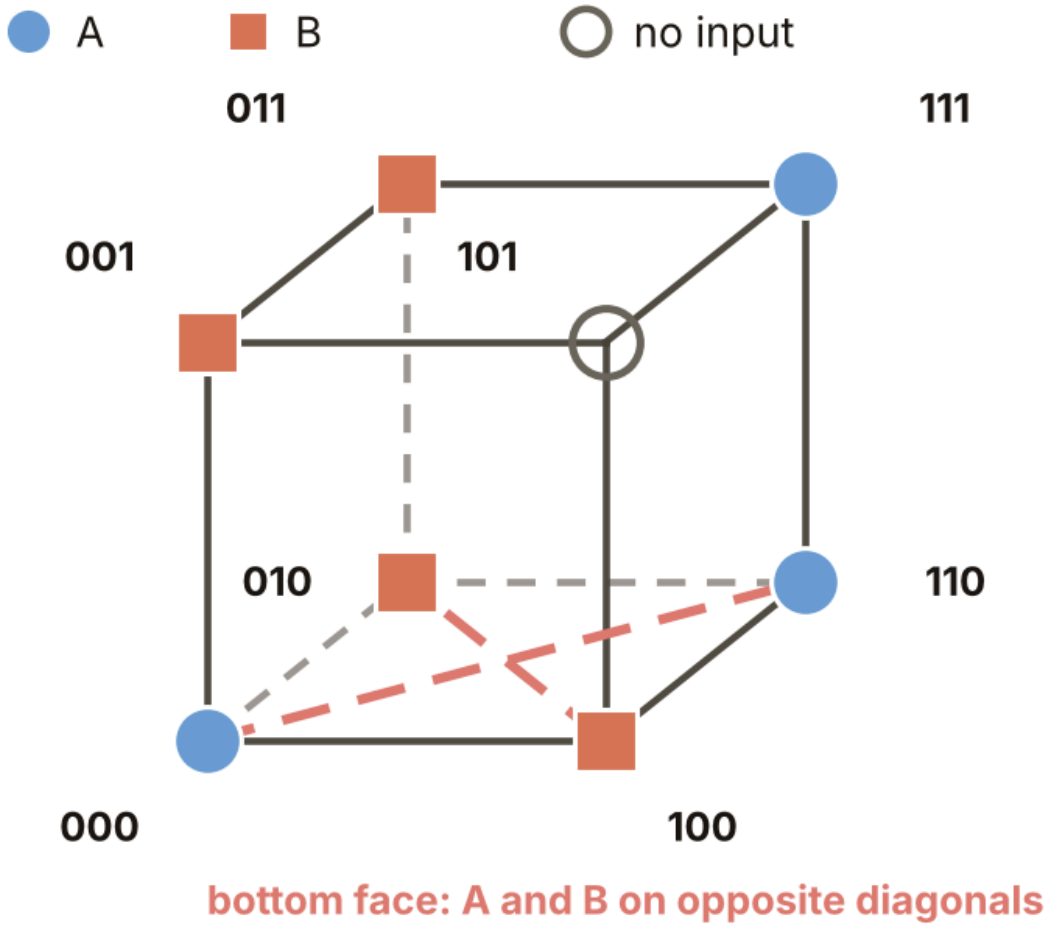
One-sentence summary: A two-layer network has a single cut on fixed codes and cannot separate corners whose labels alternate; give each target code its own detector and merge them with an OR, and any set of corners can be split into A and B.

1. The A corners are scattered, so one cut fails

Plain explanation: At the end of the previous KP, with A at 000, 110 and 111, the A and B corners on the bottom face alternate as in XOR and the last layer cannot cut them apart. The idea is not to split everything in one cut.

- **Target code**: the code of a corner labelled A, here 000, 110 and 111.

Figure `lec05-kp04-cube-target`: blue circles are A, orange squares are B, the hollow 101 is never reached by any input; the red dashed frame marks the bottom face.

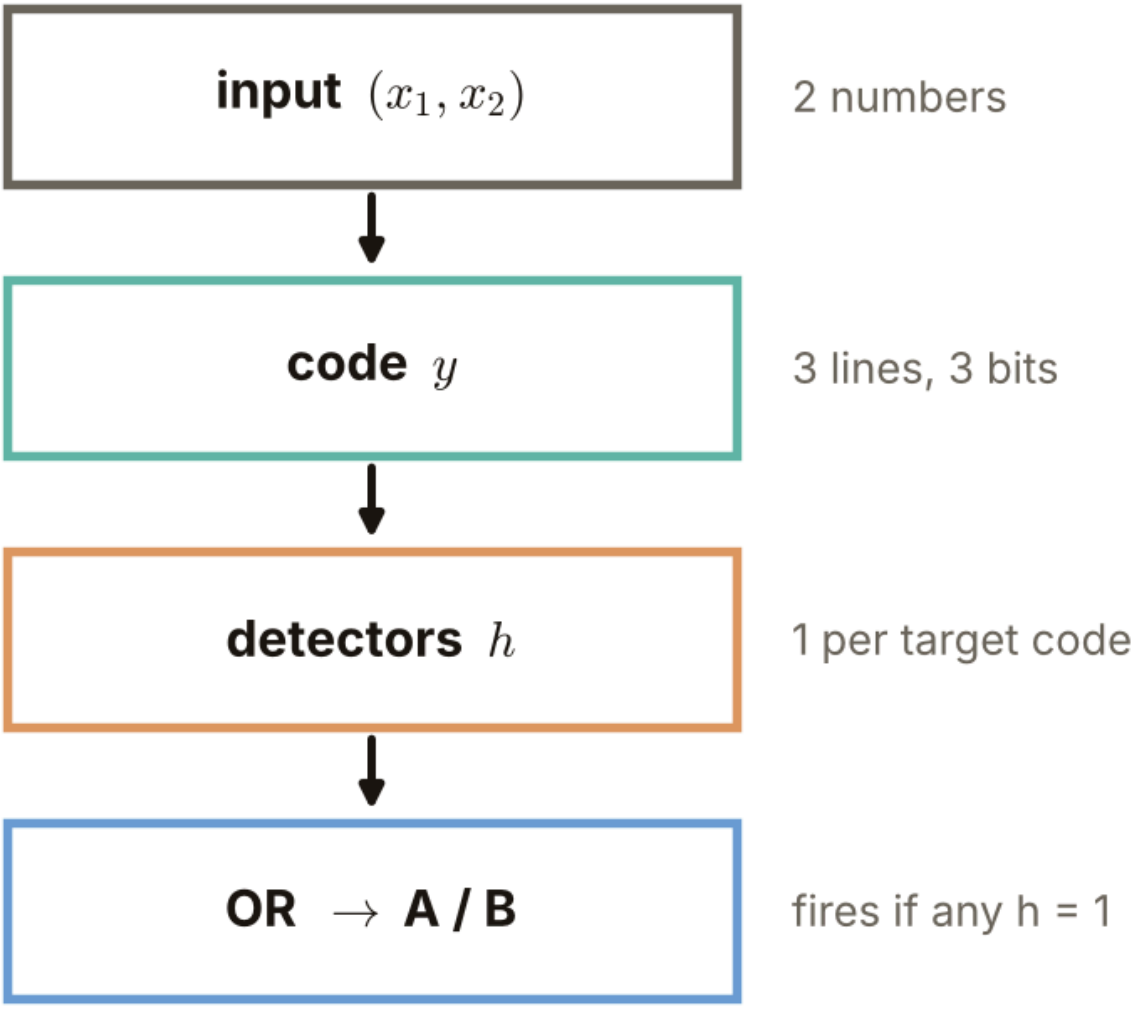


2. Three steps: code, one detector per target code, A if any detector fires

- **Three-layer network**: two hidden layers plus an output layer. The first hidden layer makes the code with three lines, the second holds the detectors and the output layer merges them.
- **Detector**: a step unit that outputs 1 exactly when the input code equals one target code, and 0 otherwise.
- **OR merge**: if any detector outputs 1 the answer is A; if all are 0 the answer is B.

The two-layer network of the previous KP had only the first layer and the output layer, without the detector layer in between.

Figure `lec05-kp04-stages`: top to bottom are the input, the code, the detectors and the OR merge.



3. Detect 110: every wrong bit costs 1 point

Formal definition: for a target code $t \in \{0,1\}^3$ with k ones, the detector is

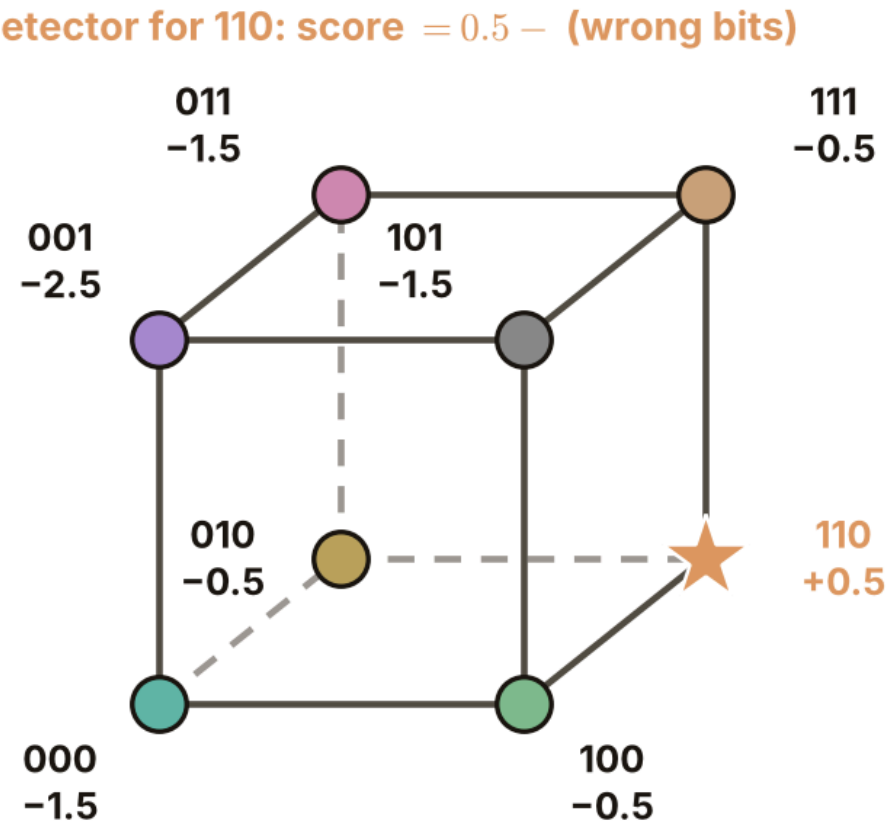
$$h_t(\mathbf{y}) = \sum_{i=1}^{3} s_i\, y_i - (k - 0.5)\,, \qquad s_i = \begin{cases} +1, & t_i = 1, \\ -1, & t_i = 0. \end{cases}$$

Step rule: $h_t > 0$ gives output 1, otherwise 0.

Why it works: $\sum_i s_i y_i = k - m$, where m is the number of bits in which $\mathbf{y}$ differs from t, so $h_t = 0.5 - m$. A perfect match gives $+0.5$, one wrong bit gives -0.5 and two give -1.5, never above 0.

Example: 110 has two ones, $h_{110} = y_1 + y_2 - y_3 - 1.5$; $y = 110$ gives 0.5 and $y = 010$ gives -0.5.

Figure `lec05-kp04-cube-one`: the number beside each corner is its score for the 110 detector; only the star is positive.

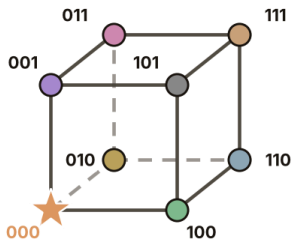


4. The same recipe gives three detectors for 000, 110 and 111

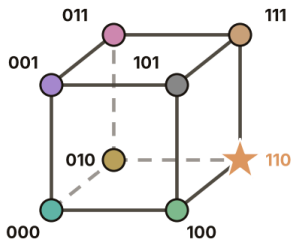
$$h_{000} = -y_1 - y_2 - y_3 + 0.5, \quad h_{110} = y_1 + y_2 - y_3 - 1.5, \quad h_{111} = y_1 + y_2 + y_3 - 2.5.$$

000 has no ones so its bias is $-(0 - 0.5) = +0.5$; 111 has three ones so its bias is -2.5. Each detector scores $+0.5$ at its own corner.

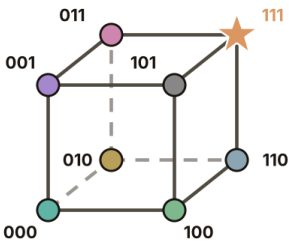
Figure `lec05-kp04-three-det`: three cubes, each marking one target code with a star.



h_{000}: detect 000



h_{110}: detect 110

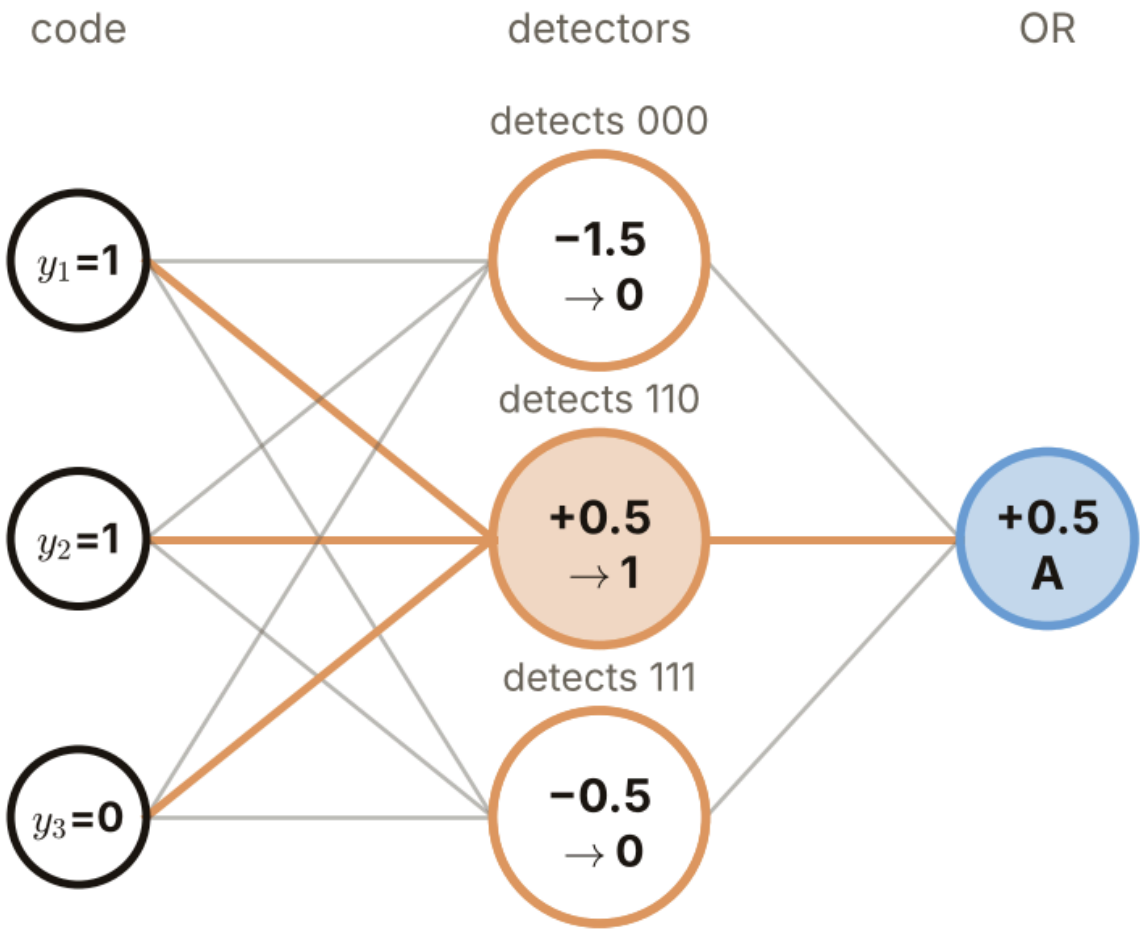


h_{111}: detect 111

5. Code 110 arrives: only the 110 detector fires

Substituting $y = (1, 1, 0)$: $h_{000} = -1.5$, $h_{110} = 0.5$, $h_{111} = -0.5$. After the step rule the three bits are $(0, 1, 0)$. The output layer adds the bits and subtracts 0.5: $0 + 1 + 0 - 0.5 = 0.5 > 0$, so A.

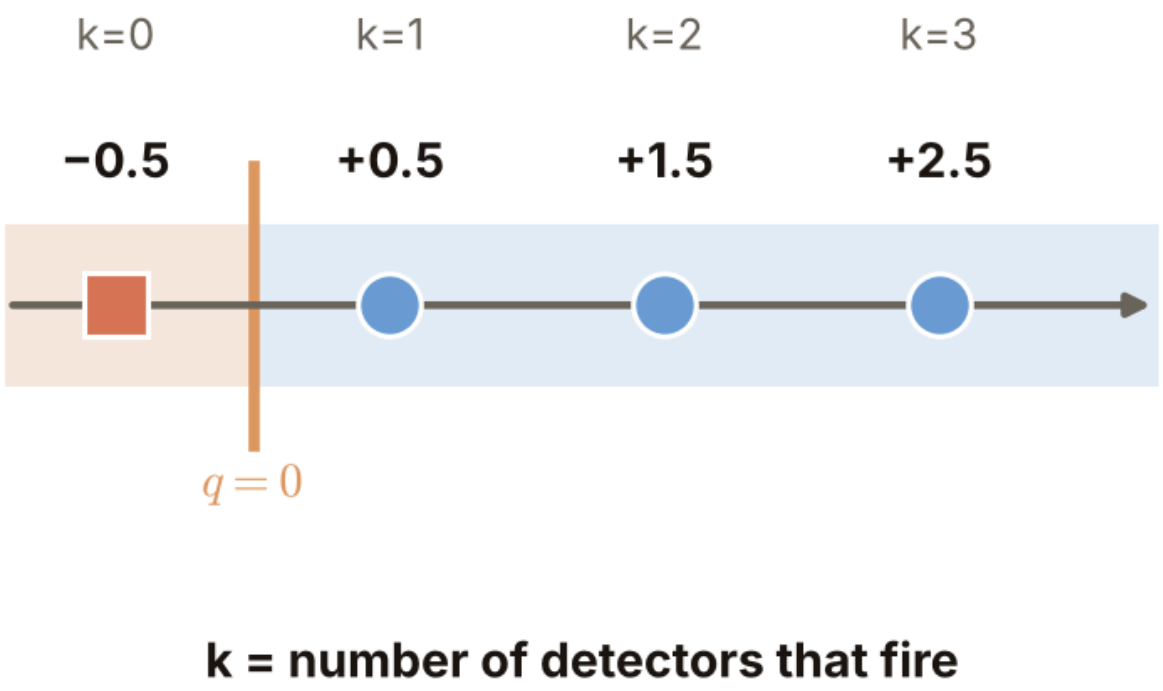
Figure `lec05-kp04-net-110`: the orange lines are the path that fires; each node shows its score and bit.



6. One rule merges them: fired count k minus 0.5, above 0 means A

The output score is $q = h_{000} + h_{110} + h_{111} - 0.5 = k - 0.5$, where k is the number of detectors that fire. $k = 0$ gives -0.5 and B; $k \geq 1$ gives $0.5,\ 1.5,\ 2.5$ and A. This is an OR merge. Every code fires at most one detector, so only $k = 0$ or 1 occurs.

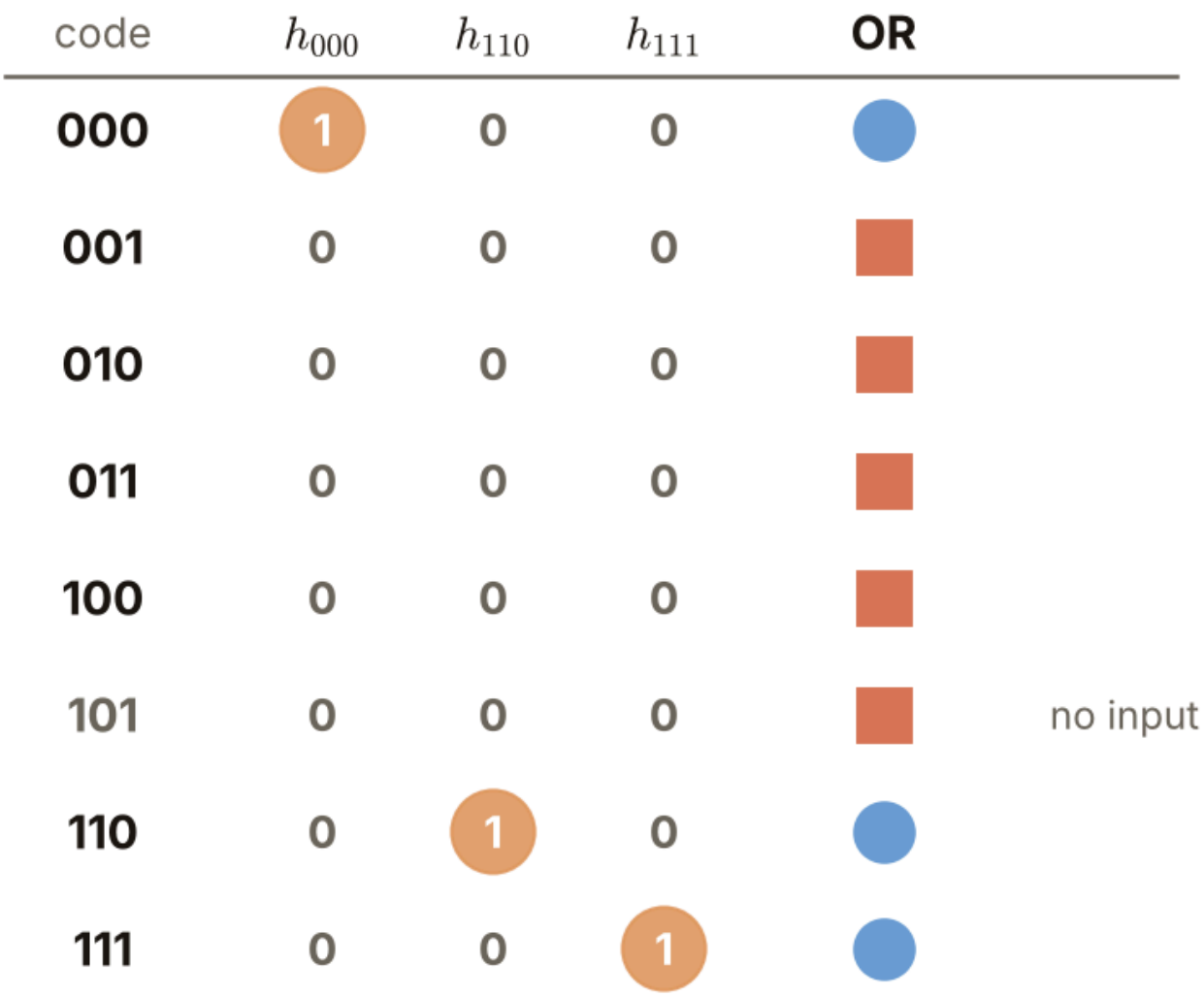
Figure `lec05-kp04-or-line`: on the number line the zero lies between $k = 0$ and $k = 1$, with B on the left and A on the right.



7. All eight corners: only 000, 110 and 111 give A

Do not check only the successful inputs. For 010 the three detector scores are $-0.5,\ -0.5,\ -1.5$, all bits are 0, the output is -0.5 and the answer is B. The unrealized code 101 also gives all zeros, so B; no input ever gets it, so it does not affect classification.

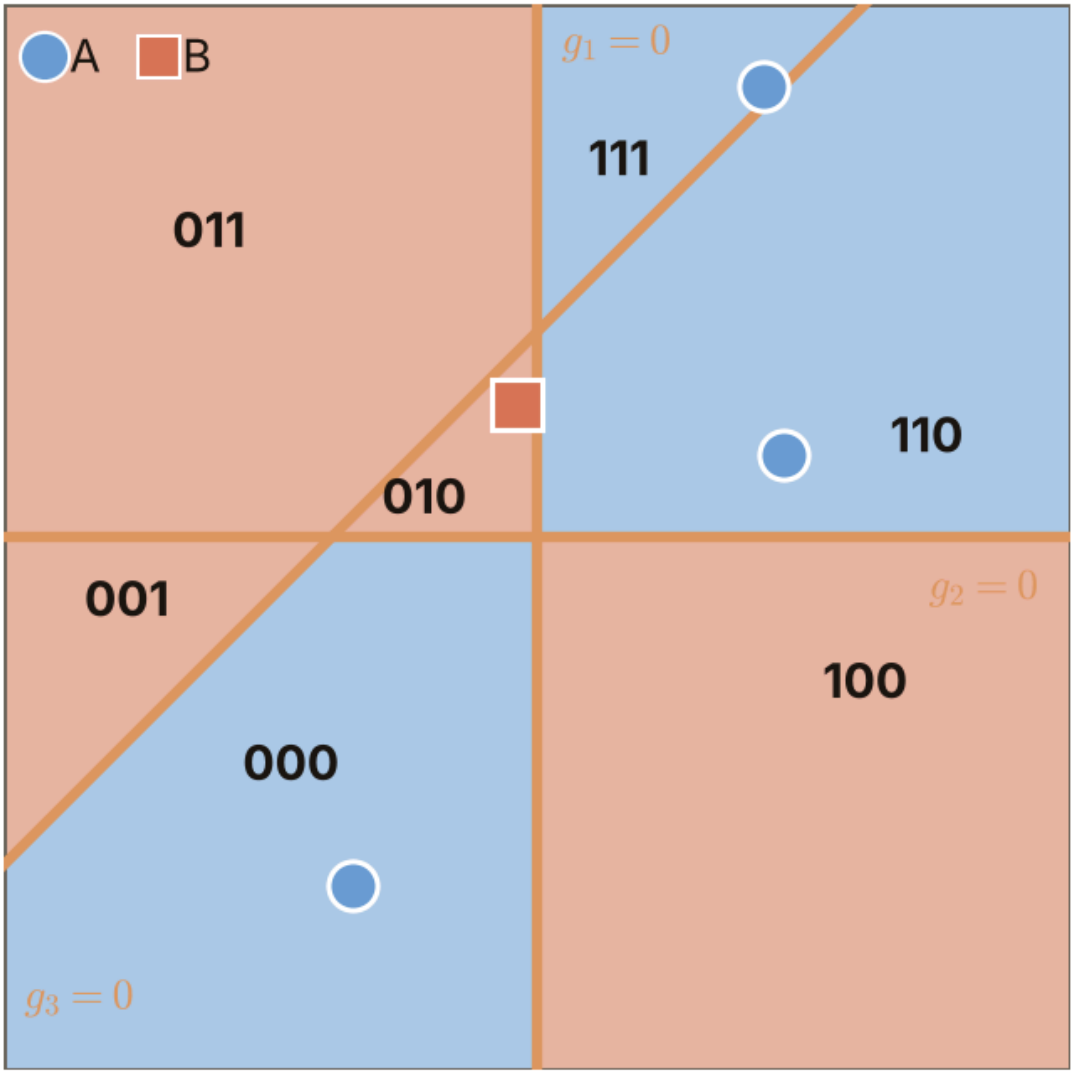
Figure `lec05-kp04-firing`: each row is a code; only 000, 110 and 111 make their own detector fire and give A.



8. Back in the input plane: the network gets all three A cells right

The A region is the union of the cells 000, 110 and 111 with the B cell 010 between them; one cut could not do this and the three-layer network does.

Figure `lec05-kp04-plane-final`: blue is the A region and orange is the B region; the A circles and the B square are classified correctly.

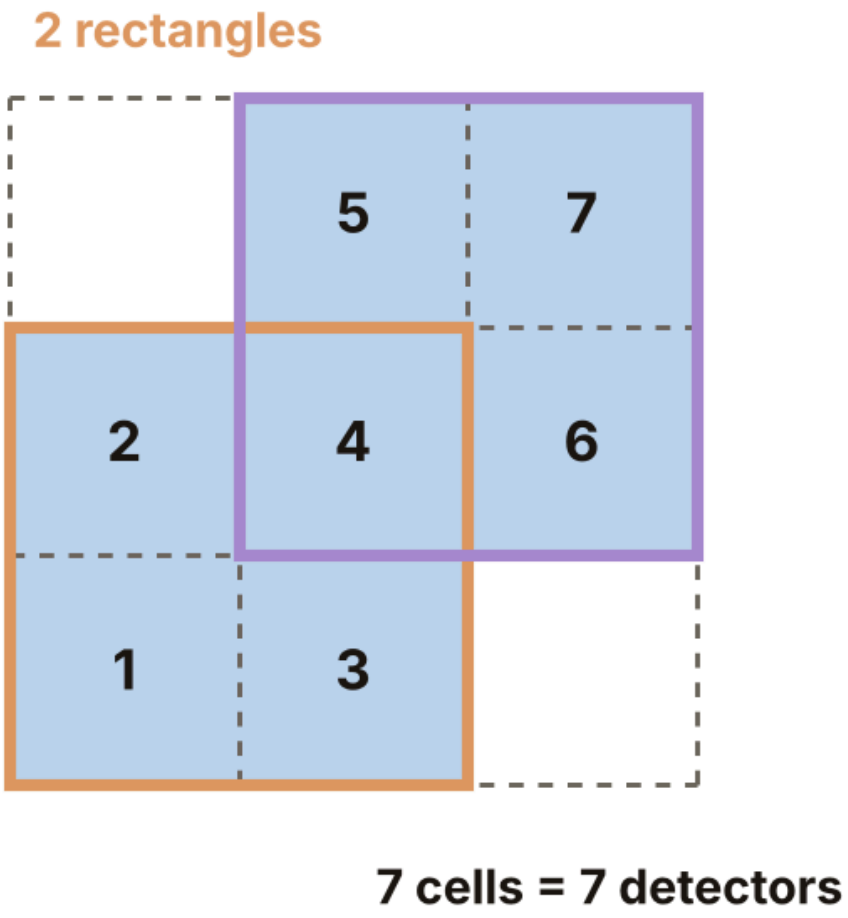


Common misconception: believing that because a three-layer construction exists, the network has learned the weights or is the smallest possible. The construction only proves that this is enough; how the weights come from labelled points is the question of the next KP.

Extension: overlapping rectangles, practice and worked solution

Overlapping rectangles [added]: two overlapping rectangles use 4 lines each, so 8 lines cut the plane into cells; A covers 7 of them and needs 7 detectors, not 2. The number of original rectangles is not the number of codes after subdivision; the recipe is unchanged.

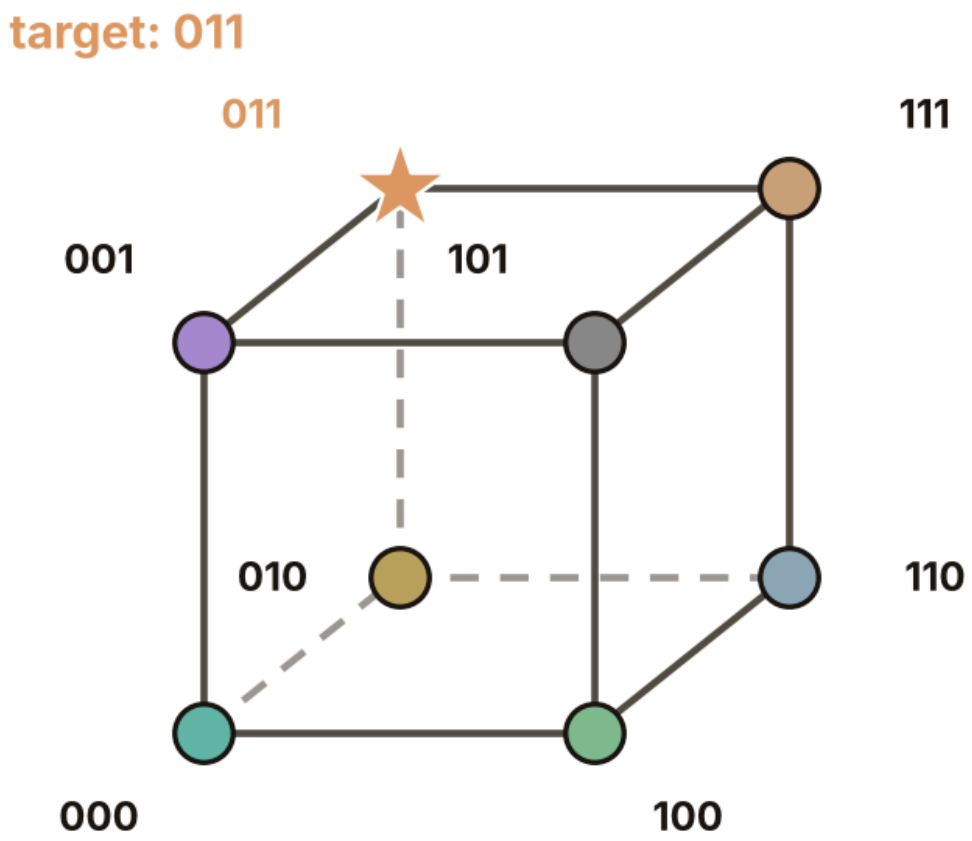
Figure `lec05-kp04-refine`: orange and purple are the two rectangles, cells numbered 1 to 7.



Practice:

1. Write the detector h_{011} for 011 and compute its score at $y = 011$ and $y = 111$. Hint: weight $+1$ where the target bit is 1, -1 where it is 0, bias $-(\text{number of ones} - 0.5)$.
2. What is the score at $y = 010$? Hint: guess first with "every wrong bit costs 1", then substitute to check.

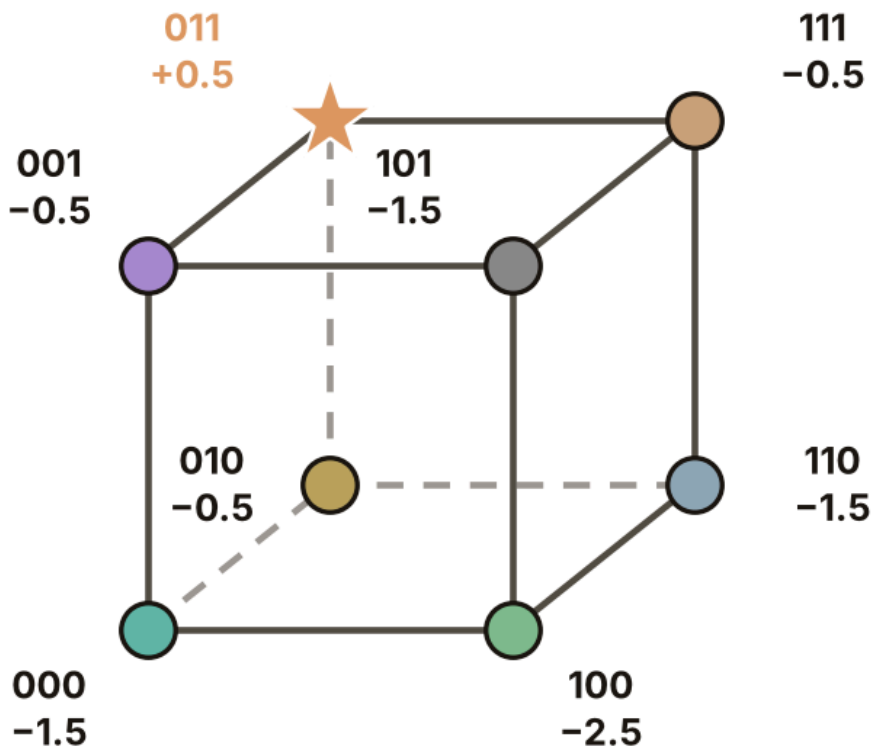
Figure `lec05-kp04-cube-011`: the star is the target code 011, scores left out.



Worked solution of questions 1 and 2: (1) Formula: 011 has two ones, the weights are $(-1, +1, +1)$ and the bias is $-(2 - 0.5) = -1.5$, so $h_{011} = -y_1 + y_2 + y_3 - 1.5$. (2) Substitute: $y = 011$: $0 + 1 + 1 - 1.5 = 0.5$; $y = 111$: $-1 + 1 + 1 - 1.5 = -0.5$. (3) Decision: 011 outputs 1 and 111 outputs 0. (4) Question 2: 010 is one bit from 011, so the score is $0.5 - 1 = -0.5$; substituting $0 + 1 + 0 - 1.5 = -0.5$ agrees.

Figure `lec05-kp04-cube-011-scores`: the number beside each corner is its score for the 011 detector.

$$h_{011} = -y_1 + y_2 + y_3 - 1.5$$



Terms and sources

English	中文
Target code	目標碼
Three-layer network	三層網路
Detector	偵測器
OR merge	或合併

Source: Chapter 4 KP04, §4.3.2 (read 2026-10-24). That a three-layer network can represent alternating corners follows the source; the general detector formula, the corner-by-corner check and the subdivision example are [added] and verified by program.